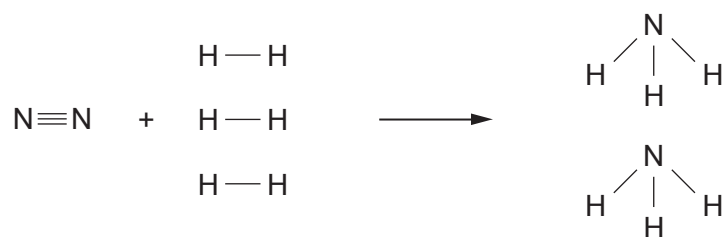


7. (a) The formation of ammonia can be represented by the following equation.



- The total amount of energy released in making the bonds in the products is 2340 kJ
- The total amount of energy released in making the bonds in the products is 94 kJ **more than** the total energy used in breaking the bonds in the reactants
- The amount of energy used to break the  $\text{N} \equiv \text{N}$  bond is 941 kJ

Use this information to calculate the energy used to break **one**  $\text{H} - \text{H}$  bond.

[3]

Energy = ..... kJ



Examiner  
only

- (b) The Haber process used to make ammonia is usually carried out at a temperature of around 400 °C. Explain why this is the optimum temperature.

[2]

A higher temperature is not used because .....

.....

A lower temperature is not used because .....

.....

- (c) Ammonia is used in the production of fertilisers such as ammonium nitrate.

Give the balanced symbol equation for the reaction between nitric acid and ammonia to form ammonium nitrate.

[2]

.....

- (d) Eutrophication is a problem caused by fertilisers being washed into waterways leading to overgrowth of plants. Describe how eutrophication leads to the death of aquatic organisms.

[2]

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| 9 |

